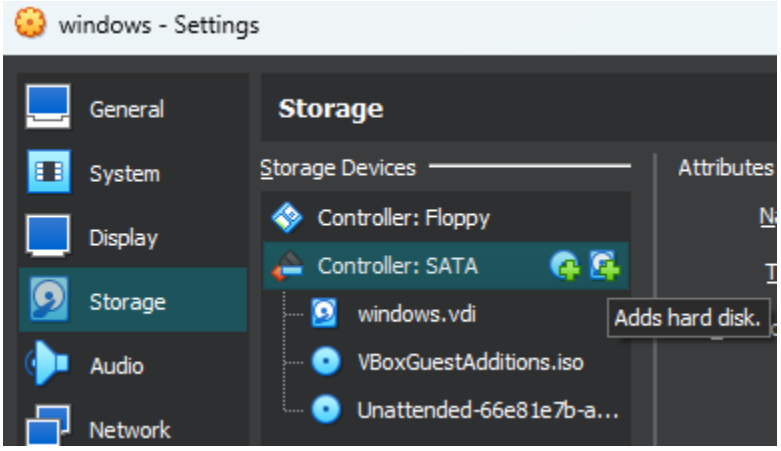
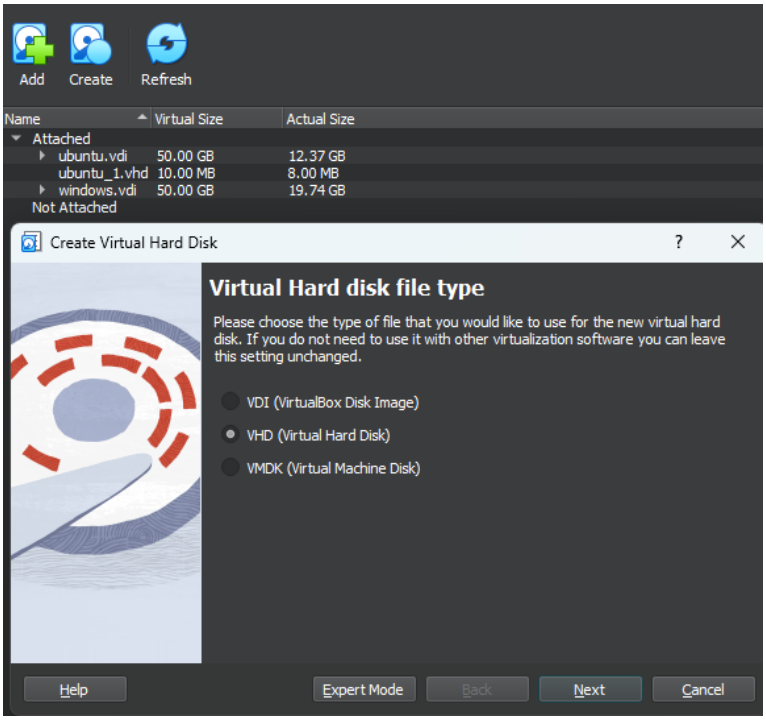
Navid Baghaei

**Lab 6**

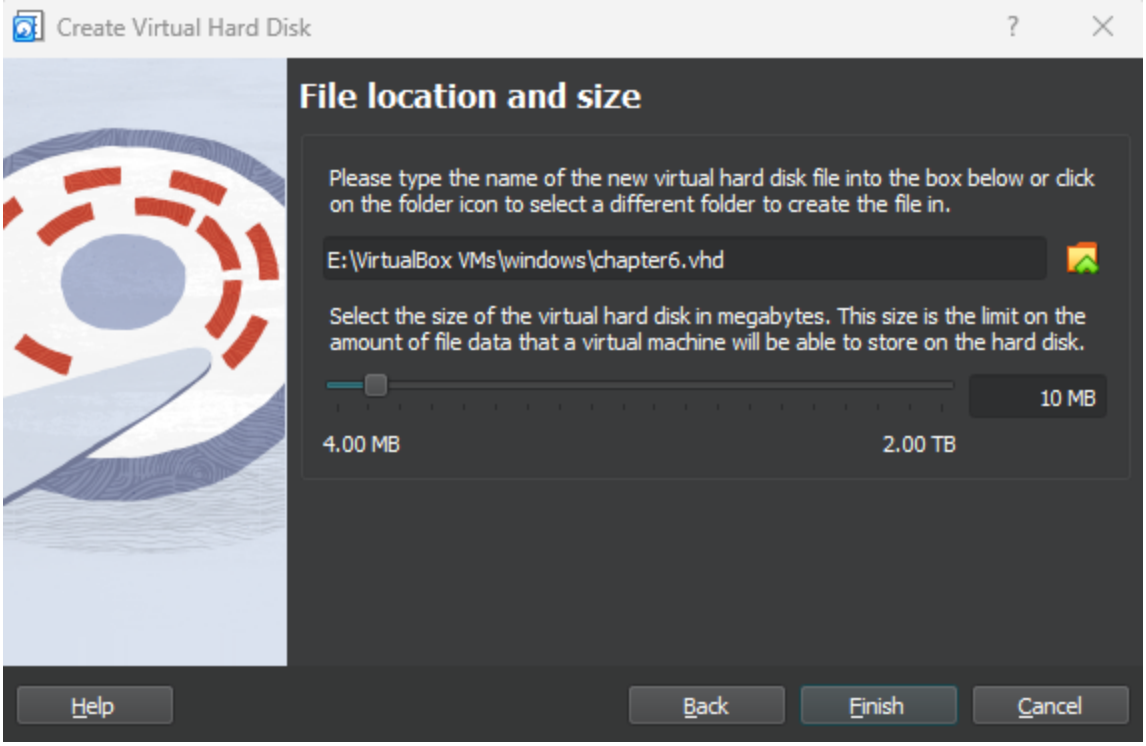
**TASK 1**

1. 

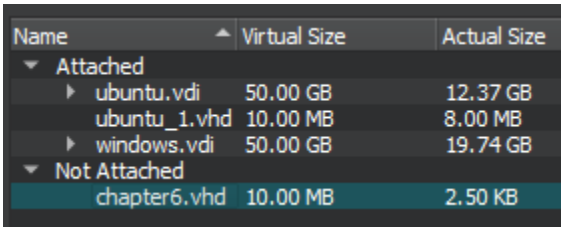

In the settings for my Windows VM, I went to storage and selected add hard disk on Controller: SATA




Selected create, and in the popup selected VHD and pressed next.

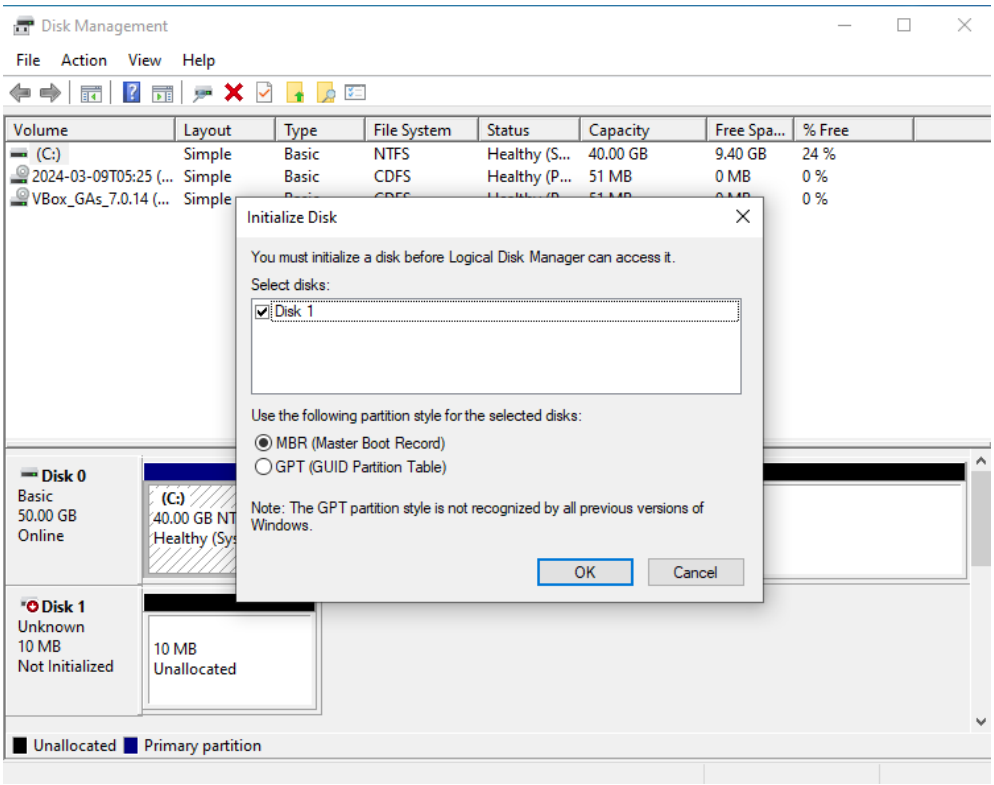



Changed the filename to chapter6.vhd and set the size to 10 MB.

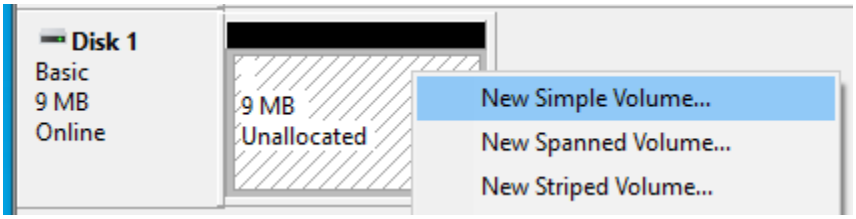


| Name | Virtual Size | Actual Size |
|---|---|---|
| Attached | | |
| ubuntu.vdi | 50.00 GB | 12.37 GB |
| ubuntu_1.vhd | 10.00 MB | 8.00 MB |
| windows.vdi | 50.00 GB | 19.74 GB |
| Not Attached | | |
| chapter6.vhd | 10.00 MB | 2.50 KB |

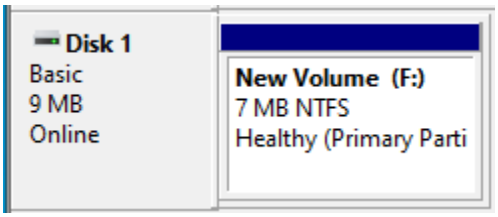
Selected the new drive and pressed choose. It is now attached.

2. 


Launched the Disk Management tool. On the popup I selected Disk 1 and checked the MBR option and pressed OK.

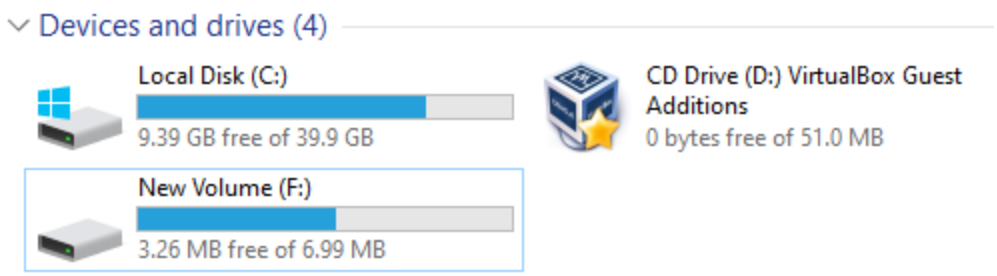



Right clicked on the disk and selected New Simple Volume

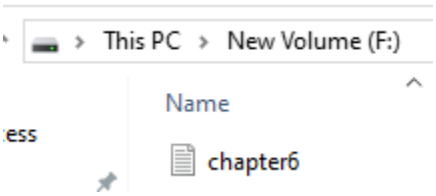



Followed the wizard and used default options. The New Volume is now set up.

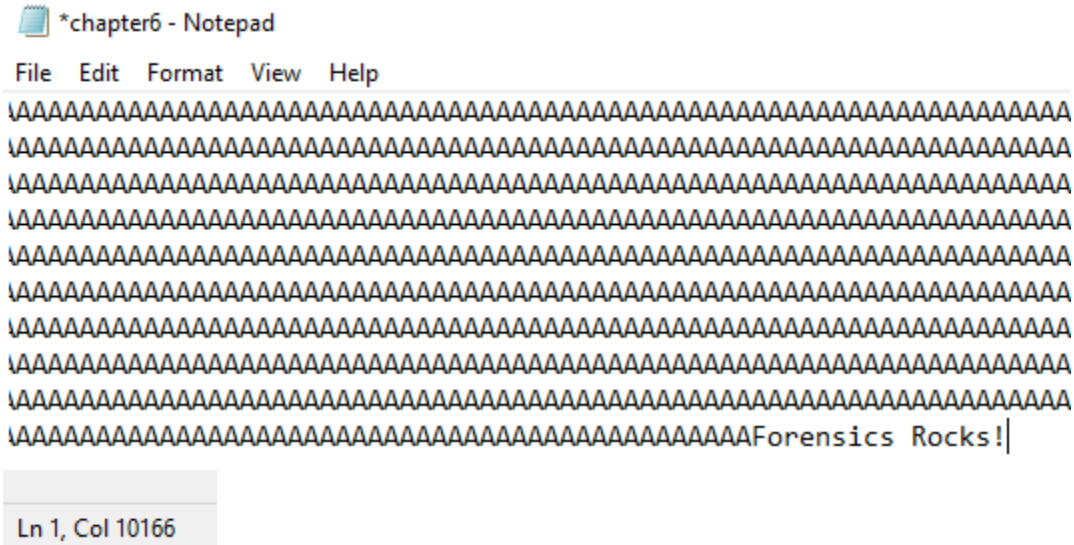
3.



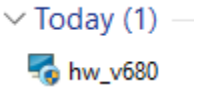

The drive shows up on file explorer.



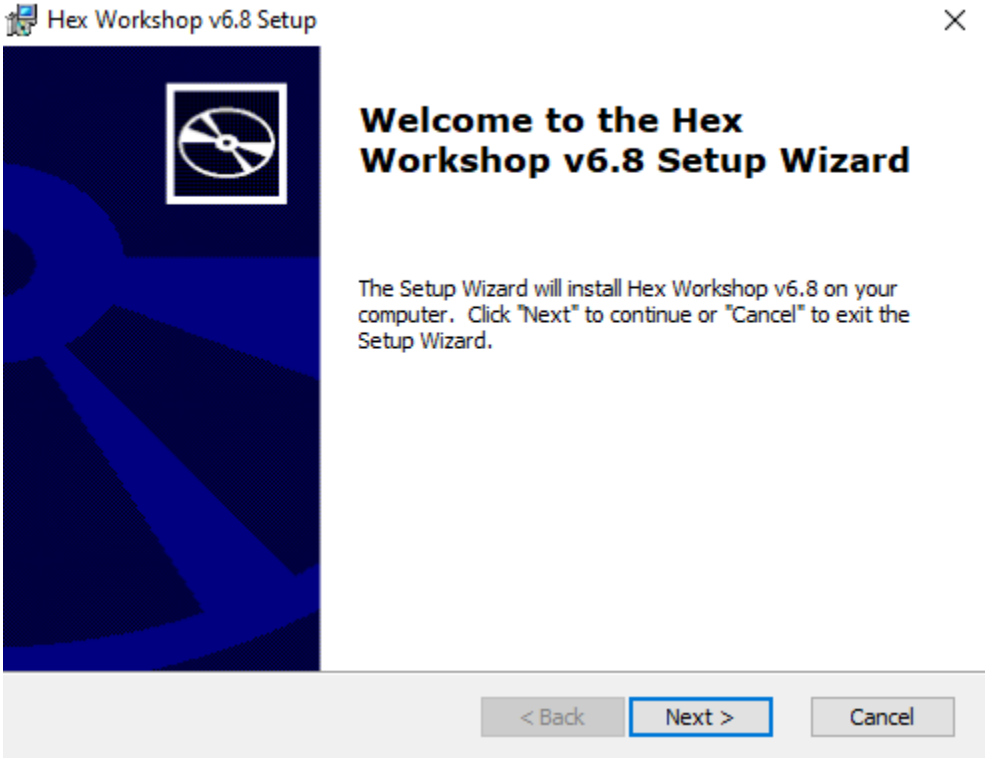

Created a text file in the new drive and named it chapter6.

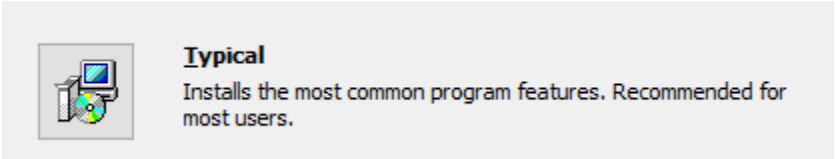

*chapter6 - Notepad
File Edit Format View Help
AAAA

4.

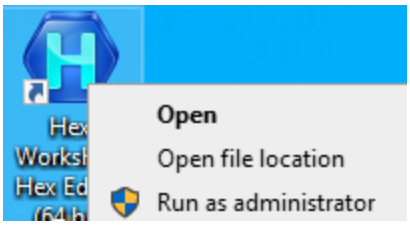



Followed the link and downloaded Hex Workshop

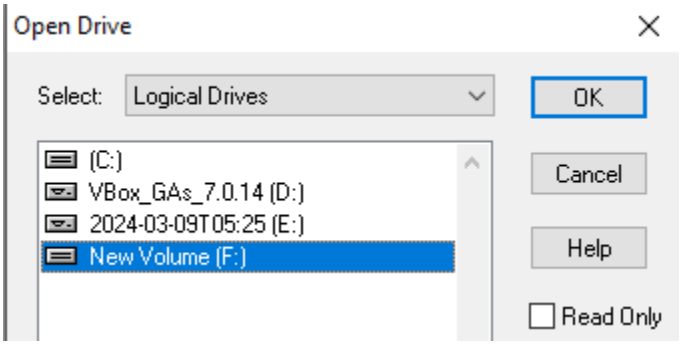


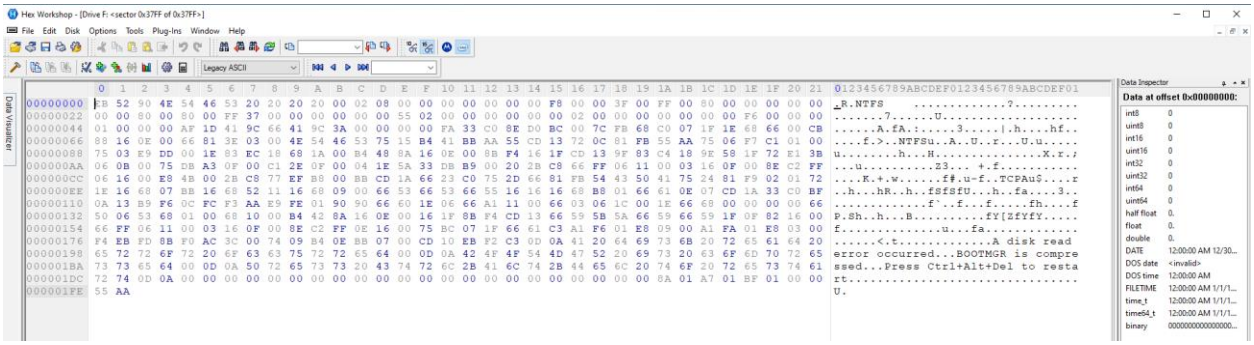

Followed the installation wizard and used the Typical option. Finished installing the program and rebooted the VM.

5. 

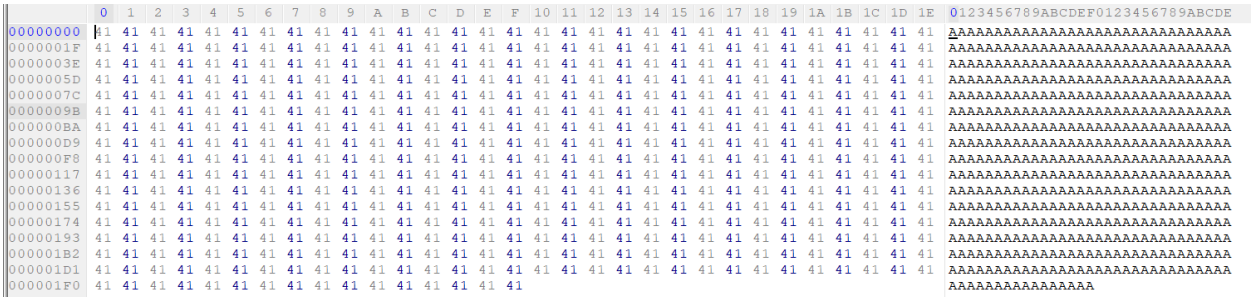

Right clicked the program and ran it as administrator.



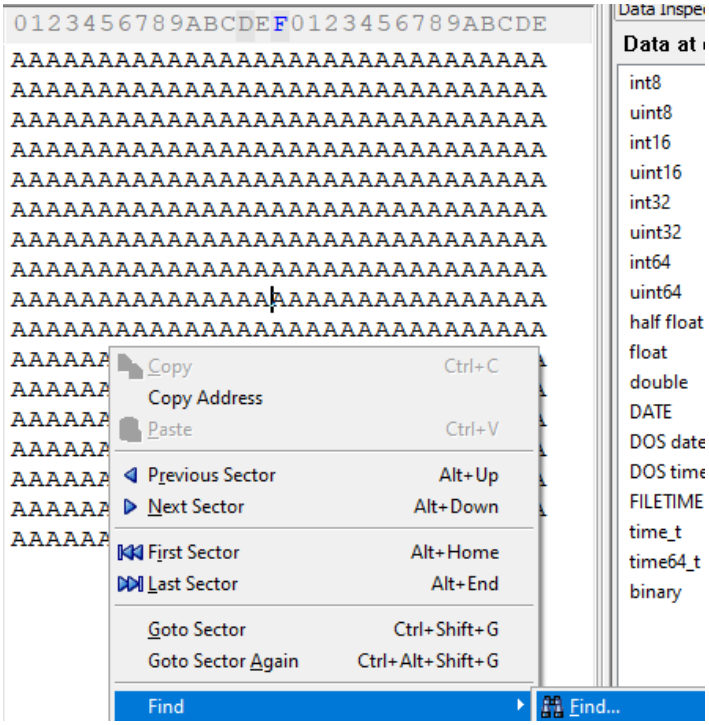

Selected Open Drive then picked the new drive made earlier.



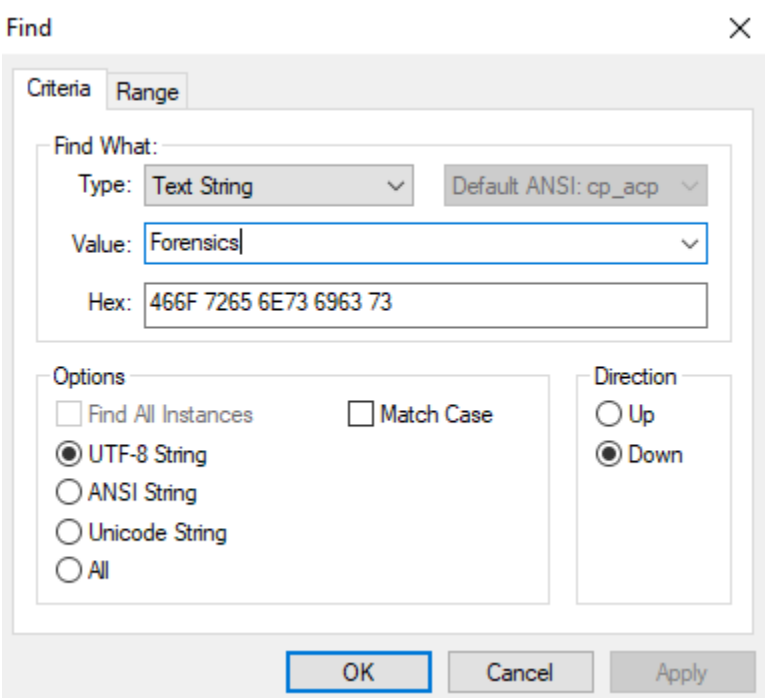

The page now shows the hexadecimals and the ASCII representations on the right.



Found the section with all of the As.




Right clicked in the section filled with As, and selected Find, then Find again.




Changed the Type to “Text String” then entered “Forensics” for the Value.

```
0123456789ABCDEF0123456789ABCDE
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAForensics Roc
ks!............................
..............................
................
```

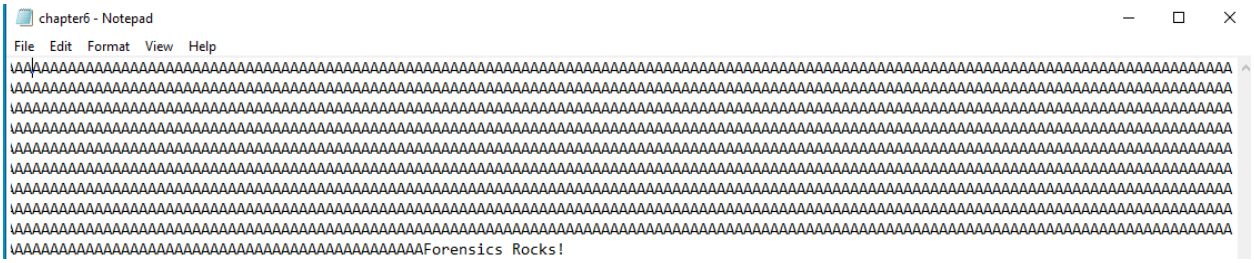
Found the Forensics Rocks text along with the file slack proceeding it. The text is red because there are unsaved changes.

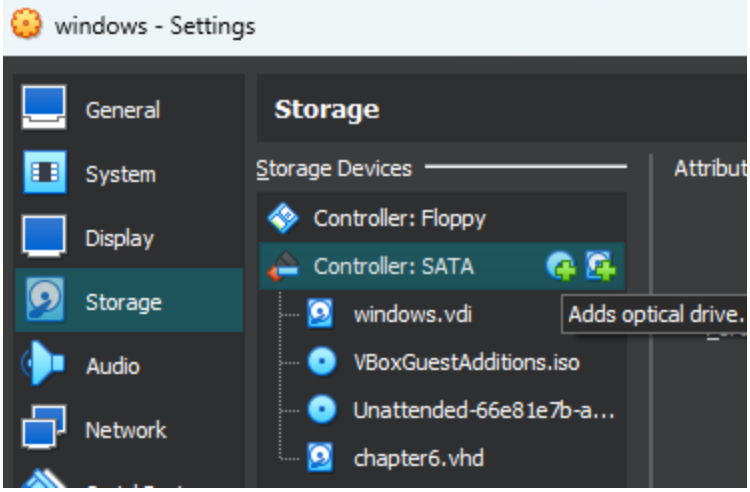
```
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAForensics Roc
ks!............................
............Secret.............
................
```

Typed the word “Secret” in the file slack space making sure to leave at least 5 characters of space to the right.

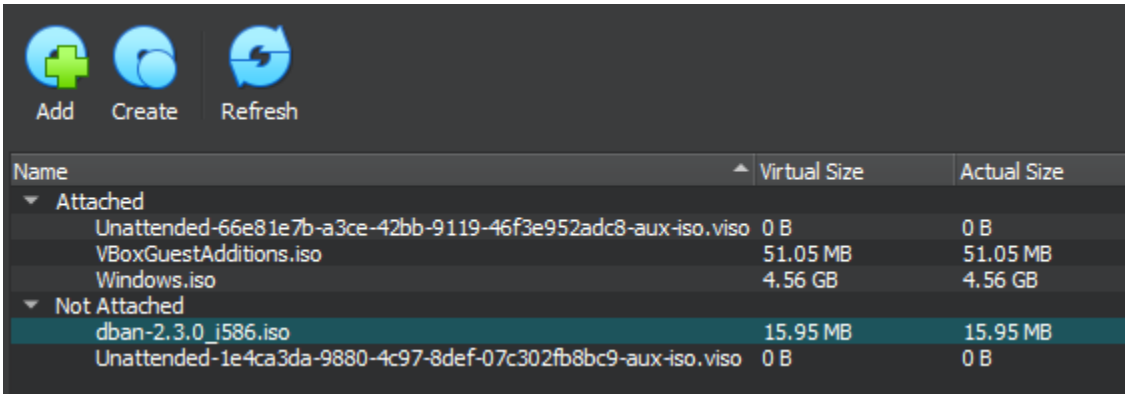
```
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAAAAForensics Roc
ks!............................
............Secret.............
................
```

After saving, the text turns to a black font.

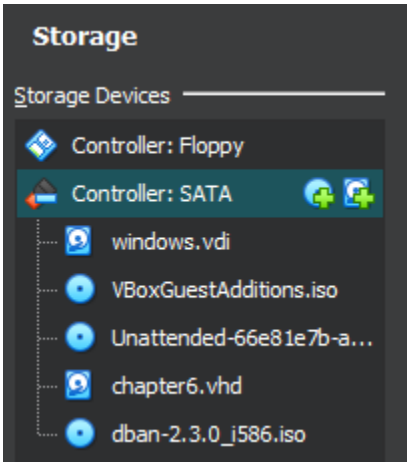

chapter6 - Notepad
File Edit Format View Help
AAAA



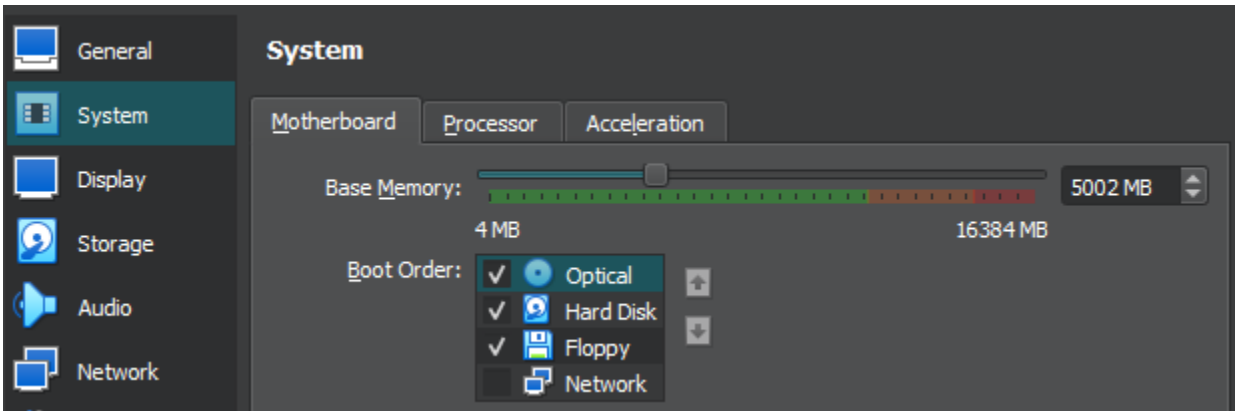

In the windows vm and under storage, I selected add optical drive for Controller: SATA.



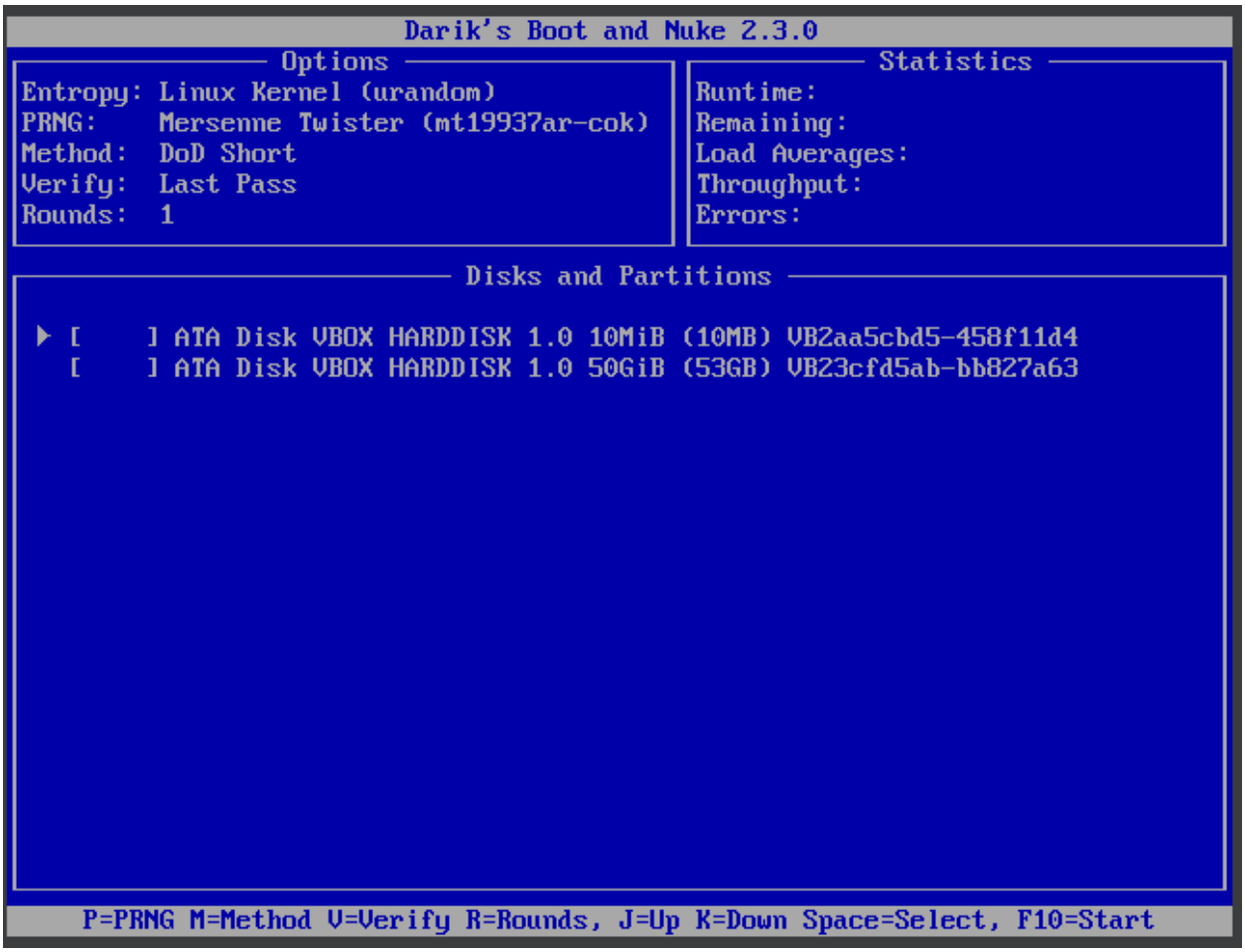

Selected Add, and picked the iso file I just downloaded on my pc. Selected it, and pressed Choose.



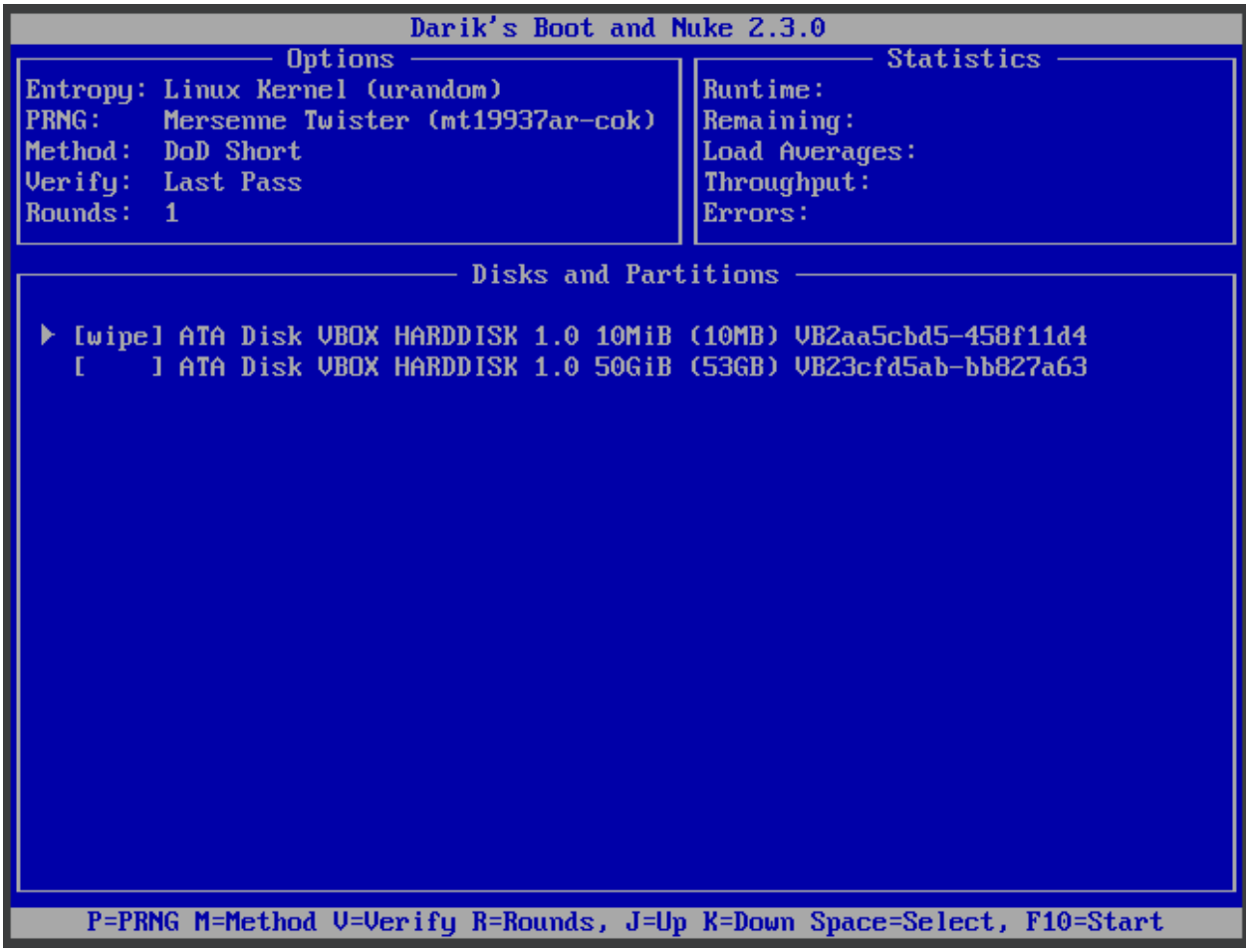

The dban now appears in the Storage Devices section.

2. 


In the windows settings, under System, “Optical” was already selected in the Boot Order. I selected it and moved it to the top of the list.

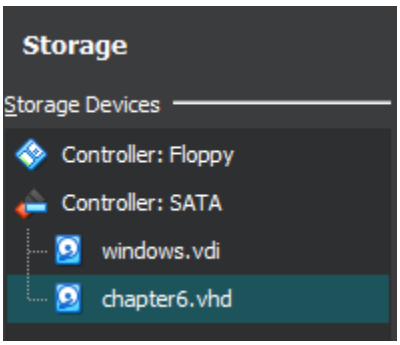



Got DBAN running. Took several hours because it was not launching for some reason, but after removing the other isos I had attached like Guest Additions, it finally ran.

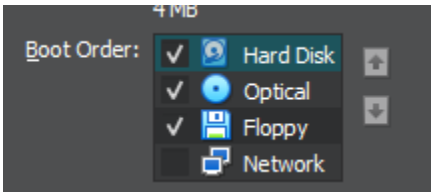



Selected the 10mb drive and pressed space to wipe it.

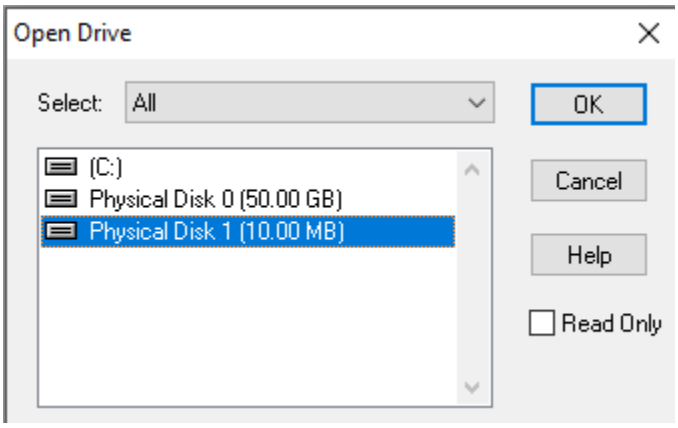
```
DBAN succeeded.
All selected disks have been wiped.
Hardware clock operation start date:  Fri Mar 08 23:55:17 2024
Hardware clock operation finish date: Fri Mar 08 23:59:08 2024

 * pass ATA Disk VBOX HARDDISK 1.0 10MiB (10MB) VB2aa5cbd5-458f11d4

USB stick(s) not detected. Unable to save logs..
Press any key to continue...
```



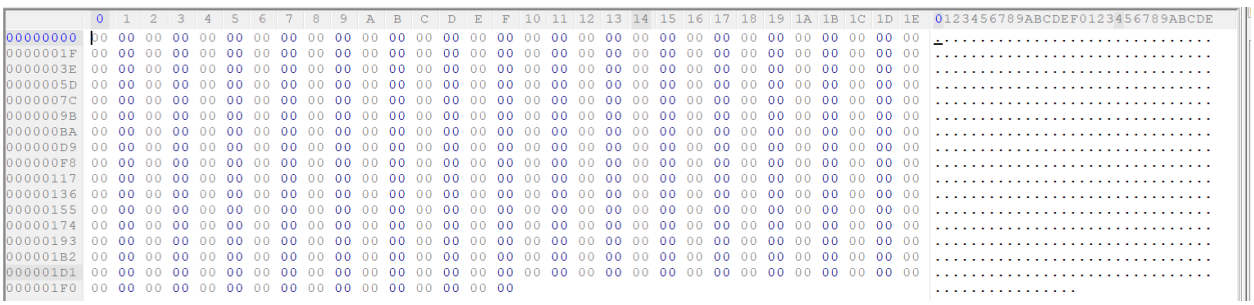

Powered off the VM, then removed the DBAN iso.




Changed the boot order back to having Hard Disk at the top.

3. 


Launched the VM again, and opened Hex Workshop. Went to Open Drive and selected the Physical Disk.



All the sectors are now empty and have no data.

**STEP 4**

1. **Gutmann Wipe**: Uses 35 different write sequences over the disk, making it extremely thorough. It employs a mix of random sequences, all zeroes, all ones, and predefined patterns.

2. **PRNG Stream**: Uses a pseudorandom number generator to overwrite data on the hard drive. It's faster than the DoD 5220.22-M or Gutmann methods but doesn't provide the same level of security. It's suitable for situations where a faster erase time is desired and the highest security isn't a priority.

3. **Autonuke**: Automatically starts a wiping process using a predefined method, simplifying the wiping process for users.

Among these, the Gutmann Wipe is considered the most secure due to its comprehensive and varied data overwriting patterns.